

Domain priors do not substitute for real operational data

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TGDC case study, research track

Every figure below is printed by `python run_all.py`.

Question

Labs generate enterprise data from procedurally built companies: an org chart, a taxonomy of work, and hand-written rules for how that work behaves. The appeal is obvious. If domain reasoning can supply the behaviour, you do not need the client's records to build a usable training set.

This paper tests whether domain reasoning can in fact supply it, for one concrete property of operational work: how long a task takes.

Hypothesis

H. Domain priors reasoned from the nature of the work capture a substantial share of what real operational records provide for predicting task duration.

The prediction was recorded before evaluation: a prior written without sight of the target data should reach an AUC of 0.60 to 0.72, clearly above the 0.5 of chance even if short of what real records achieve. Below 0.55 would mean domain reasoning supplies essentially nothing. Above 0.80 would mean it supplies nearly everything.

Why it matters. If H holds, a generator can be built for a client from their taxonomy alone. If it fails, the behaviour in generated records has to come from somewhere else, and the obvious somewhere else is the client's own data.

Method

Data. Public 311 service request feeds from New York, Chicago and Austin. These are real operational records, written by the organisations that do the work. Each carries an opening time, a service category, an owning department, an intake channel and a closing time. We take two complete calendar weeks from each city, the weeks of 4 May and 8 June 2026.

city	train	test	median resolution
New York	72,206	78,296	5.4h
Chicago	17,291	38,671	118.0h
Austin	5,968	6,029	23.8h

Tickets still open at the pull date are kept and treated as slow, since all are months past any threshold used here. Dropping them removes the slowest work. Chicago closes most of its feed within a second of creation, so those informational records are excluded and only worked tickets remain.

Task. At ticket opening, predict whether it will take longer than the median to resolve. The model sees only what is known at that moment. Logistic regression on one-hot categoricals throughout, so the learner is held constant and differences between arms come from the training data. Recovered skill is the share of the ceiling's advantage over chance that an arm reaches, $(\text{arm} - 0.5)/(\text{ceiling} - 0.5)$.

Two information conditions. The independent variable is what the author of the prior could see when writing it.

Blind. Rules for Austin, written from Austin's service type and department vocabulary alone. The taxonomy was pulled with the query restricted to category names and row counts, so no duration, rate or per-category outcome was available. The rules were written, committed with the recorded prediction above, and only then evaluated. This is the condition H is about.

Informed. Rules for New York, written with that city's conditional resolution rates available. This is the condition a lab is in when it builds a generator while inspecting a client's data, and it bounds what the same procedure achieves with site statistics in hand.

Carried across. The New York rules, unmodified, applied to Chicago. This separates rules that encode general domain structure from rules that encode one site.

The pre-registration is checkable. The blind rules are commit 8973439 and their evaluation is commit 4400226.

Results

The blind prior reaches chance

prior	tested on	had the rates	AUC [95% CI]	ceiling	recov.	vs chance
Austin, pre-reg.	Austin	no	0.494 [0.481, 0.508]	0.886	−1%	indistinguishable
New York	Chicago	no	0.496 [0.492, 0.499]	0.729	−2%	below
New York	New York	yes	0.840 [0.837, 0.843]	0.918	81%	above

Intervals are 1,000-resample bootstraps of the test set. The null baseline, always predicting the majority class, scores 0.500 by construction.

H is refuted. The blind prior scores 0.494, and its interval spans chance. It is not weakly informative, it is statistically indistinguishable from guessing. The recorded prediction of 0.60 to 0.72 was wrong, and wrong in the direction that says domain reasoning supplied nothing at all.

This is not a coverage failure. The Austin rules fired on 98.9% of tickets, producing scores spread from 0.05 to 0.95. Tickets they called slow ran slow 60.1% of the time and tickets they called fast ran slow 52.9% of the time, a gap of seven points that does not survive as ranking skill.

The same procedure reaches 81% of the ceiling when the author has the site’s rates in hand. The distance between 81% and nothing is the measure of how much of that performance comes from the site rather than from the domain.

The carried-across arm makes the same point from the other direction, and slightly more sharply: its interval sits entirely below 0.5, so on Chicago the New York rules are not merely uninformative but actively misleading. Their department rules match 0.0% of Chicago rows, being New York acronyms, and their keyword list is New York housing-stock vocabulary that in Chicago points the wrong way. Rows it calls slow are slow 47.3% of the time against 58.6% for everything else.

The columns were not the problem

A prior that reasons over service category and department could in principle reach whatever those two columns support. Fitting the conditional rate directly from training records, using nothing but those same two columns, gives the ceiling for any rule of that shape:

city	best from category and department	the prior reached	gap
Austin (blind)	0.883	0.494	0.389
New York (informed)	0.918	0.840	0.078

The columns carry nearly all of the available signal in both cities. What separates the two rows is whether the author had the rates. The gap is the price of not knowing them, and it is five times larger when the author is working blind.

Why domain reasoning fails here

The blind prior rests on four claims about the nature of municipal work. One person attending once is fast. An inspection that opens a process is slow. Work needing a crew and materials is slow. A fixed-cycle routine service is fast. Each is plausible, and each is wrong somewhere.

service type	predicted slow	actually slow
ARR Compost	5%	96%
Vehicle Abatement Report	5%	94%
ARR Bulk	5%	92%
Animal Protection, Loose Dog	5%	81%
Traffic Signal Maintenance	90%	8%
Parking Violation Enforcement	30%	0%
Request Code Officer	95%	82%

A traffic signal is physically crew work but administratively a same-day safety priority. Compost collection is physically routine, but the ticket stays open across the collection cycle. Vehicle abatement is a police matter subject to a statutory waiting period.

In each case the duration is set by the organisation’s workflow and its closing conventions, not by the nature of the job. That is the answer to the question: what real operational records supply here is institutional convention, and convention is local and arbitrary enough that reasoning about the work does not reach it.

Distribution fidelity does not detect the problem

A generator ladder shows why a lab would not notice. Trained on generated New York records and scored on real ones:

generator	AUC	recovered	records that cannot occur
real records (ceiling)	0.918	100%	0.0%
marginal only	0.466	−8%	70.5%
marginals + pairwise	0.917	100%	0.0%
empirical joint	0.917	100%	0.0%

Matching marginals and pairwise dependence, the bar that SDV’s Quality Score and the NIST differential-privacy winners optimise, recovers everything on this task and invents no impossible records. The New York task is close to a lookup on one column: the service category alone scores 0.916 against the 0.918 ceiling, and the single rule **department is not NYPD** scores 0.850. An evaluation built on this task will certify almost anything that gets one column roughly right.

A second recorded prediction also failed. We expected generated records violating real-world constraints to explain downstream loss. Marginal-only generation does invent impossible department and category pairs, 70.5% of its records, and it does lose everything. But the standard bar invents none and loses nothing, so at the fidelity order that matters there was nothing to measure.

Limits

One author wrote both priors by one procedure. A practitioner with real municipal operations experience might write rules that reach further, and nothing here separates the procedure from the person. This is the largest limit: the result bounds what this procedure achieves, not what domain reasoning achieves in principle.

Three cities, one domain, one task family, one model class. Municipal service records may be unusually institution-bound. A domain with genuinely standardised process, such as a regulated industry or a franchise operation, could behave differently, and that is the first thing we would test next.

Chicago’s June week ran roughly twice the volume of its May week, so the carried-across arm compares two operating conditions as well as two cities. Its ceiling of 0.729 is also the lowest of the three, leaving less skill available to recover.

We tried a harder framing, predicting speed within a service category to remove the dominant column. Its ceiling is 0.514, too close to chance to separate anything, so we report it as a limit and not a result.

New York carries known artefacts, including one department that closes 89.7% of its tickets at exactly midnight, so its recorded durations are administrative.

That many high-order joints share low-order marginals, and that low-order fidelity therefore fails to imply downstream utility, is established. We include the ladder because it explains why a distribution-level check misses this.

What it would take to answer the general question

Several independent authors writing priors under the blind condition would separate the procedure from the person, which is the limit that most constrains this result. Domains chosen to include genuinely standardised process would establish whether institutional locality is a property of operational data in general or of public services in particular.

The procedural point stands on its own. A prior can only be evaluated as a prior once per dataset, before anyone has seen the outcomes, and the recorded prediction has to come first. That is the only arrangement here that produced a number worth trusting, and it cost an afternoon.